

Supplementary Materials for C²GES: A Diagnostic Evaluation of Role-Conditioned Extractive Summarization for Long Power-System Technical Reports

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Contents and Data Files

This supplement separates scientific interpretation from implementation records. It contains twenty tables and two figures referenced by the main manuscript. The main-text evidence-layer table (“Evidence layers and claim bounds”) states the four evaluation layers and what each may and may not support; the tables below are implementation and exploratory-component records for those layers, not a confirmatory external test. Paths below are relative to the C2GES release root. Machine-readable files are provided under `03_Reproducibility/Data/`: the non-verbatim sampling frames, aggregate metrics, paired ROUGE-L differences, output-length records, exact sign-flip results, page locators, component-factorial diagnostics, and development-only calibration report. Source PDFs and verbatim extracted passages are excluded because third-party redistribution permission has not been established.

Table S1. Rights-Safe Sampling Frame

The complete 40-row sampling frame is supplied as `03_Reproducibility/Data/rights_safe_metadata/rights_safe_report_metadata.csv`. Its 23 fields include the report identifier, source locator, split, series identifier, inclusion decision, source-file hash, and redistribution status; it contains no report passages.

Quantity	Value
Inspected complete PDFs	40
Declared pages	3200
Retained / excluded reports	27 / 13
Development / test reports	12 / 15
Non-summary candidates	12,924

Table S2. Output-Length Records

The complete 14 condition-budget summary is supplied as `03_Reproducibility/Data/postrun_diagnostics/output_length_summary.csv`; per-report values are in `03_Reproducibility/Data/postrun_diagnostics/output_length_per_report.csv`. “Unrenorm. removal” denotes the unrenormalized coefficient-removal comparison.

<i>K</i>	Condition	Mean words	Mean chars.	> 100	Table	Max. words
5	Full C ² GES	287.7	1843.6	12	14	140
5	Unrenorm. removal	329.0	2099.9	15	16	270
5	Semantic-MMR	184.7	1242.1	5	9	138
5	TextRank	177.0	1125.1	1	7	124
10	Full C ² GES	568.9	3619.5	25	26	270
10	Unrenorm. removal	596.9	3822.5	28	26	270
10	Semantic-MMR	354.5	2361.2	7	16	138
10	TextRank	369.0	2329.7	4	18	130

Table S3. Corpus and Implementation Checks

Checked property	Recorded result
Reference/body page-interval overlap	0
Normalized exact candidate/reference matches	0
Normalized common substrings of at least 50 characters	0
Report-condition-budget records / expected records	210 / 210
Selected source references resolved to a valid page	1575 / 1575
Maximum path expansions / configured limit	19,638 / 2,000,000
Maximum qualifying paths / configured limit	16,182 / 250,000
Full/unrenormalized-removal score comparisons changed	9774 / 19,008
Report-budget selection sequences changed	28 / 30
Aggregate and paired-contrast recomputation	Exact agreement

Aggregate values are available at `03_Reproducibility/Data/audits/aggregate_metrics.json`; non-verbatim paired differences and page locators are available at `03_Reproducibility/Data/figure_inputs/paired_rougel_differences_nonverbatim.csv` and `03_Reproducibility/Data/postrun_diagnostics/selected_page_locator.csv`. Detailed checks are retained with the corresponding code and archived run records. Passing these checks establishes consistency of the reported computation; it does not determine semantic summary quality.

Table S4. Exact Paired Sign-Flip Sensitivity

For each contrast, all $2^{15} = 32,768$ report-level sign assignments were enumerated. The procedure assumes report-level independence and sign exchangeability. Holm adjustment treats the six values as one family.

K	Baseline	Signs + / - / 0	Exact p	Holm-adjusted p
5	Unrenorm. removal	7/7/1	0.436768	0.436768
5	Semantic-MMR	14/1/0	0.000305	0.001526
5	TextRank	14/1/0	0.000122	0.000732
10	Unrenorm. removal	6/8/1	0.200684	0.401367
10	Semantic-MMR	11/4/0	0.006592	0.026367
10	TextRank	13/2/0	0.009644	0.028931

These post-result values are sensitivity analyses. The main manuscript emphasizes effect sizes, paired distributions, and report-composition intervals.

The complete machine-readable result is `03_Reproducibility/Data/postrun_sensitivity/exact_signflip_results.csv`; the JSON file in the same directory retains the report-level non-verbatim deltas used in the calculation.

Table S5. Post-Access External-Series Inventory

The fixed exploratory inventory contained eight public power-system reports from four organizations. Seven PDFs were acquired and analyzed; the NESO item returned HTTP 403 and was retained as a documented acquisition failure rather than replaced. Metadata or search snippets had been seen before the protocol was frozen, so all rows are classified as exploratory and none permits a confirmatory claim.

Organization	Fixed	Analyzed	Evidence status
ENTSO-E	3	3	Post-access exploratory
NERC	3	3	Post-access exploratory
NESO	1	0	Acquisition failed (HTTP 403)
Transpower	1	1	Post-access exploratory
Total	8	7	Confirmatory claims disallowed

The non-verbatim record, source pages, direct-download locators, access date, and redistribution status are supplied in `03_Reproducibility/Data/exploratory_external_v0/EXPLORATORY_EXTERNAL_INVENTORY.csv`.

Table S6. Controlled Comparator and Budget Conditions

Control	Frozen implementation	Diagnostic purpose
Candidate pool	One layout-filtered pool per report for every method	Removes candidate-access asymmetry
Output budget	Complete-ranking greedy selection at 110 and 260 words; no sentence truncation	Matches emitted information quantity
Development opportunity	Nine configurations for each tunable system in the exploratory registry	Balances search opportunity
Series unit	Seven reports in seven distinct series	Supports equal-series contrasts and LOSO checks
Component scale	Positive weights renormalized to sum to one	Separates component removal from score-scale change
Failure policy	Every fixed item retained; execution status saved per cell	Prevents silent replacement or deletion

The v3 run completed all 182 condition–report–budget cells. These controls were fixed after source access and therefore improve internal comparability without converting the pilot into a prospective confirmatory experiment.

Table S7. Principal Component Contrasts

Family	Contrast and budget	Mean Δ R-L	95% cluster interval	Exact p	Holm p
Incremental	AB-5 – AB-6, 110	-0.00799	[-0.01949, 0.00149]	0.34375	1.00000
Incremental	AB-5 – AB-6, 260	-0.00718	[-0.01855, 0.00151]	0.28125	1.00000
Graph type	G-T – G-U, 110	0.00139	[-0.01481, 0.01959]	0.90625	0.90625
Graph type	G-T – G-U, 260	-0.00544	[-0.01640, 0.00594]	0.35938	0.71875
Factorial	Reservation main, 110	-0.00971	[-0.01751, -0.00244]	0.04688	0.23438
Factorial	Path main, 110	-0.00247	[-0.01787, 0.01333]	0.79688	1.00000
Factorial	Interaction, 110	-0.01104	[-0.03173, 0.00264]	0.34375	1.00000
Factorial	Reservation main, 260	0.00047	[-0.00387, 0.00530]	0.87500	1.00000
Factorial	Path main, 260	-0.01034	[-0.01871, -0.00371]	0.03125	0.18750
Factorial	Interaction, 260	0.00632	[-0.00222, 0.01602]	0.25000	1.00000

No component contrast survived its separately defined Holm family. Intervals use series-cluster resampling, and exact values enumerate the 2^7 series sign assignments. Full series effects, standardized paired effects, and leave-one-series-out estimates are provided in `factorial_series_effects.csv` and `factorial_loso.csv` under the v3 factorial directory.

Table S8. Complete AB-0–AB-6 Component Chain

Budget	Condition	R-L	Redundancy	Role cov.	Edge cov.	Words/util.	Time (s)/MB
110	AB-0	0.1564	0.2308	0.286	0.662	108.4/0.986	0.269/1.04
110	AB-1	0.1582	0.1527	0.524	0.869	108.4/0.986	0.268/1.04
110	AB-2	0.1696	0.0783	1.000	0.890	108.1/0.983	0.253/1.03
110	AB-3	0.1707	0.0800	1.000	0.971	108.1/0.983	0.248/1.03
110	AB-4	0.1577	0.0564	1.000	0.964	109.0/0.991	0.249/1.03
110	AB-5	0.1588	0.0508	1.000	0.964	108.6/0.987	0.252/1.06
110	AB-6	0.1668	0.0595	1.000	0.948	108.4/0.986	0.252/1.08
260	AB-0	0.1687	0.1799	0.429	0.626	258.0/0.992	0.350/1.12
260	AB-1	0.1873	0.1185	0.714	0.885	259.3/0.997	0.347/1.12
260	AB-2	0.1843	0.1128	1.000	0.906	258.6/0.995	0.305/1.12
260	AB-3	0.1848	0.1038	1.000	0.971	257.7/0.991	0.314/1.12
260	AB-4	0.1805	0.0915	1.000	1.000	257.9/0.992	0.313/1.12
260	AB-5	0.1772	0.0509	1.000	1.000	258.1/0.993	0.344/1.45
260	AB-6	0.1844	0.0747	1.000	0.984	257.4/0.990	0.342/1.46

All failure rates were zero. Runtime is the recorded per-report estimate and memory is Python peak allocation; the separately loaded embedding model is excluded. Role and edge coverage measure heuristic formal coverage, not human-validated semantic correctness.

Table S9. Adjacent-Corpus Construct Audit of Role and Edge Cues

Two public corpora carrying human annotation were used to bound what the lexical role and edge channels can carry. The deterministic role rule and its transition rule were applied unchanged; neither corpus is redistributed, and both must be obtained from their original distributors. This is an adjacent-corpus construct audit, not the domain expert adjudication reserved in the main text.

Corpus	Endpoint	Units	Human positive rate	Precision	Recall	Note
GUM eRST	Dominant role assigned	7678 EDUs	—	—	—	349 units (4.55%)
GUM eRST	Admissible typed edge	7285 pairs	0.179	not estimable	not estimable	45 pairs (0.62%) had role evidence on both units
EBM-NLP	interventions → mitigation	2139 sentences	0.524	0.610	0.064	382 sentences (17.9%) received a dominant role
EBM-NLP	outcomes → impact	2139 sentences	0.519	0.807	0.060	same sentence set as above

The GUM rows use the human enhanced-RST relation labels of the Georgetown University Multilayer corpus; the EBM-NLP rows use test spans annotated by medical professionals, with two roles mapped onto C2GES role names (intervention to mitigation, outcome to impact) and the participant label reported as unmapped. The cues are precise when they fire and sparse and genre-bound in general text; neither row closes any precondition of the annotated-study gate in the main text, and no row supports a role, edge, path, faithfulness, or omission claim.

Table S10. External Prospective Evaluation on a New Report Family

The inventory, reference rule, budgets, comparators, endpoints and statistics were frozen before any score was computed (protocol and amendments in 03_Reproducibility/Data/external_prospective_v1/). The corpus was acquired on 22–23 September 2026 from the public document stores of ENTSO-E and from additional NERC reports; it is a frozen external corpus, not an author-attested unseen set, and it is never pooled with the historical 15-report test or the seven-series pilot.

Selector	n	Mean ROUGE-L @110	Mean ROUGE-L @260	Word budget
no-path C ² GES	19	0.125803	0.141786	110 / 260
Full C ² GES (path 0.10)	19	0.123381	0.138880	110 / 260
TextRank	19	0.149076	0.164726	110 / 260

Contrast	Budget	Mean Δ R-L	Bootstrap 95%	Exact p	Holm p	Negative / positive documents
Full – no-path	110	-0.002422	[-0.005913, 0.000310]	0.164062	0.317284	5 / 3
Full – no-path	260	-0.002906	[-0.005605, -0.000636]	0.015625	0.046875	7 / 0
TextRank – no-path	110	0.023273	[-0.005924, 0.055280]	0.158642	0.317284	6 / 13
TextRank – no-path	260	0.022940	[0.008540, 0.038543]	0.008095	0.032380	5 / 14

Composition: six ENTSO-E grid incidents (including the 2024 South-East Europe and 2025 Iberia reports), seven ENTSO-E market-coupling incidents, and six NERC reports; 21–2,000 candidate units and 102–10,935 reference words per document. References are the official summary sections detected after converting each PDF to structured Markdown; the detector accepts Executive Summary, Management Summary and Summary headings, which is a documented generalisation of the earlier Executive-Summary rule made before scoring. Holm adjustment covers the four contrasts shown. The two corrected contrasts favour the untyped ranking over the typed path channel; they do not establish system superiority, expert construct validity, or operational benefit, and they are not author-attested unseen evidence.

Post-freeze extension (sensitivity, not a pre-registered test). A later acquisition added five documents (three grid incidents and two frequency-deviation reports), raising the corpus to 24. The direction is unchanged and slightly larger for the path channel — Full minus no-path is -0.006908 at 260 words with 10 negative, 1 positive and 13 exactly tied documents — but mean-based inference is not stable: the randomised sign-flip value rises to 0.444 (Holm 0.678) because one document with a large positive difference dominates the mean, and the TextRank contrast falls to 0.130 (Holm 0.390). Reporting the extension is therefore a statement about the fragility of mean-based tests on a zero-inflated, heavy-tailed paired distribution, not a replacement of the frozen 19-document result.

Arm-level decomposition (sensitivity). Testing the two layers separately rather than jointly shows the same budget dependence. Adding role evidence and role-group reservation to a lexical-and-position baseline changes mean ROUGE-L by -0.03510 at 110 words (18 of 22 discordant documents negative; $p = 0.0039$, Holm 0.0774), by -0.01776 at 260 words, and by $+0.00626$ and $+0.01079$ under five- and ten-unit budgets; none reaches family-wise correction. The path channel stays at or below zero in all four budgets. The role layer’s apparent benefit is therefore tied to the equal-unit-count protocol, whereas the negative component result is stable across budgets and corpora.

Table S11. Error Profile of Path-Driven Selection Changes

Descriptive decomposition of the units whose selection changed between Full (AB-5) and no-path (AB-6) in the seven-series matched-budget pilot, grouped by the sign of the report-level ROUGE-L difference. Changed units are the symmetric difference of the two selection sets in a report–budget cell; cells are counted when at least one unit changed. “Blocked pairs” count ordered role-compatible pairs (a permitted stage-monotone transition) whose positions differ by more than the 12-position window and therefore could not form an edge. Chronology-violating incident edges are typed edges whose source unit occurs later in the document than its target; direction follows the role stage, not document order, so a high share is a structural property of the design rather than an error count. Machine-readable per-unit records and inputs are in `03_Reproducibility/Data/descriptive_addenda_v1/`. This profile describes where the path channel acts; it does not apportion causality for the endpoint difference.

Budget	Δ R-L sign	Cells	Changed units	Edge-incident	Chronology-violating incident	Mean max incident distance	Mean blocked pairs
110	negative	4	21	0.952	0.800	9.9	83.3
110	zero	1	5	1.000	1.000	12.0	417.8
110	positive	2	11	0.818	1.000	9.9	33.4
110	all	7	37	0.919	0.882	10.2	113.7
260	negative	4	27	0.963	0.885	10.5	50.0
260	positive	2	22	1.000	0.909	11.5	362.8
260	all	6	49	0.980	0.896	11.0	190.4

At both budgets the units the path channel moves are almost always edge-incident (92–98%), and most of those touch at least one chronology-violating edge (80–100%). The endpoint-negative cells do not show an exceptional profile: at 110 words they exhibit fewer blocked beyond-window pairs per changed unit than the zero/positive cells, and at 260 words fewer than the positive cells. The 12-position window leaves a large stock of role-compatible pairs unreachable (tens to hundreds per changed unit), so long-range dependencies remain untested by design rather than adjudicated by evidence.

Table S12. Per-Role Token Alignment Between Extracts and References

Descriptive cue-versus-cue alignment for the role layer in the same pilot. Both sides are labeled by the pipeline’s deterministic lexical cues (unit-scaled within each text set, abstaining on ties): a selected unit “has” a role when its dominant role equals that label, and reference sentences are labeled the same way after a deterministic sentence split. Token precision is the share of stopword-filtered content tokens of the role- r selected units that also occur in role- r reference sentences; token coverage is the share of role- r reference-sentence tokens recovered anywhere in the extract. Sentence-level ROUGE-L matching was evaluated first and essentially never fires, because executive-summary sentences are longer and differently worded than source units; token alignment is the informative granularity. Reference-side roles are sparse (root cause never fires on any of the seven references), so empty cells are structural, not missing data. These are alignment measures under the cue definition, not expert-validated correctness, and they close no row of the evidence gates.

Role	Ref. reports	110-word budget				260-word budget			
		Precision		Coverage		Precision		Coverage	
		Full	no-path	Full	no-path	Full	no-path	Full	no-path
Root cause	0/7	0	0	—	—	0	0	—	—
Trigger event	1/7	0	0	0.053	0.053	0.008	0	0.105	0.158
Propagation or response	4/7	0.251	0.106	0.217	0.243	0.213	0.084	0.397	0.369
Impact	4/7	0.093	0.075	0.144	0.072	0.050	0.040	0.252	0.249
Mitigation	2/7	0.024	0.061	0.125	0.188	0.061	0.061	0.225	0.275

Macro means over the seven reports; empty cells mark roles that never fire on that reference. Precision reads low in absolute terms because the same content is expressed with different words on the two sides; the relative pattern is the informative part. The role layer’s effect is concentrated where the cues and the references coincide at all (propagation/response, impact, mitigation), and the path channel does not systematically raise alignment: Full exceeds no-path in propagation precision at both budgets but trails it in mitigation precision at 110 words and in mitigation coverage at both budgets.

Table S13. Long-Block Audit Detail

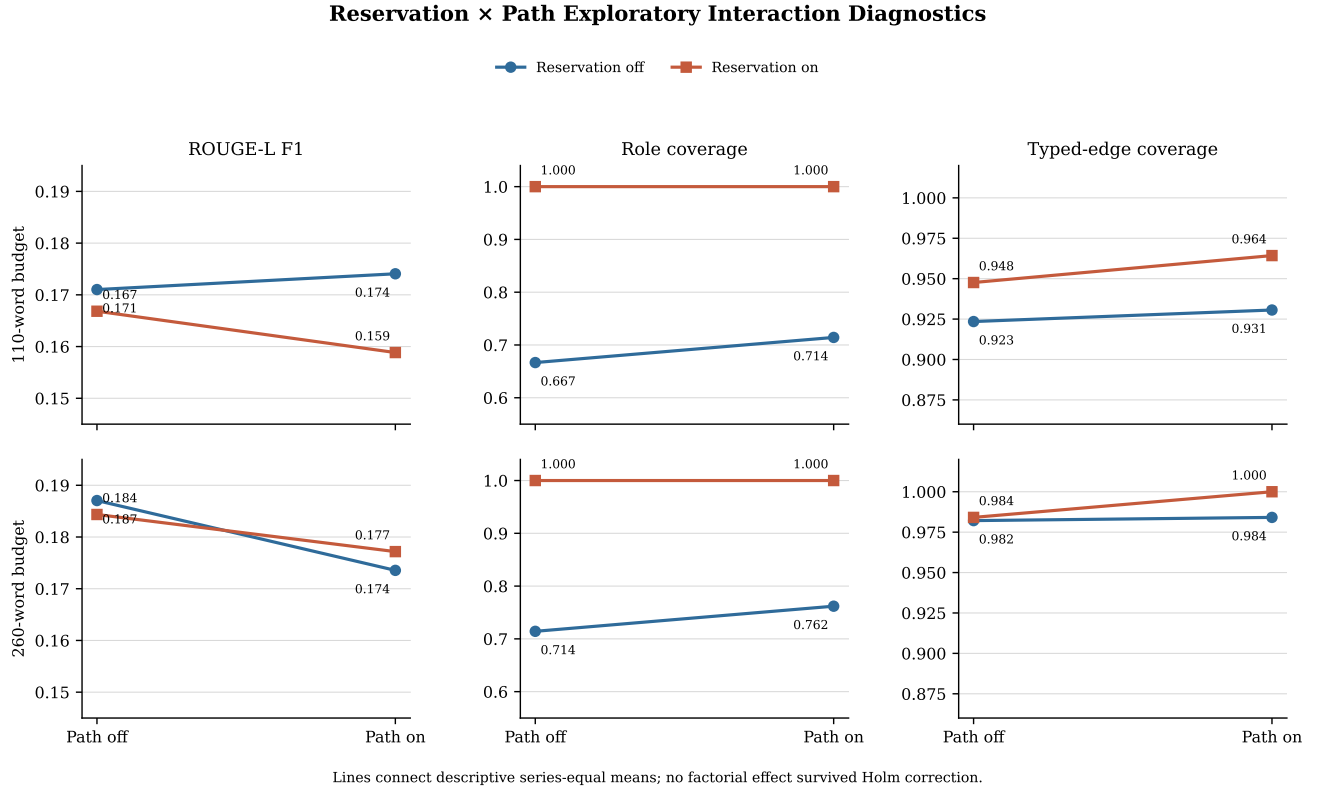
Per-condition counts of long extraction units in the legacy candidate stream (unit budgets $K=5$ and $K=10$, 15 historical test reports; columns sum selection instances, so a unit selected under several conditions is counted in each) and the block-preserving re-segmentation audit over all 27 historical reports. The audit preserved page/block boundaries, excluded recurrent margin material, and dropped blocks overlapping detected tables; it was not used to rerun ranking, so whether the relative ordering at matched budgets would change remains unanswered by design.

Condition	$K=5$			$K=10$		
	Units >100 words	Table-marker units	Max unit words	Units >100 words	Table-marker units	Max unit words
Lead	5	2	119	6	7	119
Centroid	11	15	221	13	29	221
TextRank	1	7	124	4	18	130
Semantic-MMR	5	9	138	7	16	138
Role	3	6	270	6	12	270
Graph, no path (strict)	15	16	270	28	26	270
Full C ² GES	12	14	140	25	26	270

Under the block-preserving rule the same reports yield 14,290 layout units of which 39 exceed 100 words (spread over 16 of 27 reports); the per-report maximum unit length has median 103 and worst case 522 words. Non-verbatim examples of long legacy selections are recorded in `q10_long_unit_examples_nonverbatim.csv` within the same data directory: the longest Full selection is a 270-word unit sharing 21 content tokens with its reference summary, and a 162-word Full selection shares 40. Long units can therefore carry substantial reference-token mass into an extract, which is the mechanism behind

the output-length diagnostic in the main text; the audit quantifies the exposure but does not license re-ranking claims.

Figure S4. Reservation–Path Factorial Interaction



Lines compare reservation off/on while the horizontal axis switches the normalized path term. The rows show 110- and 260-word budgets; columns show ROUGE-L, formal role coverage, and formal typed-edge coverage. The budget-dependent ROUGE-L patterns do not establish a stable interaction after family-wise correction.

Table S14. Reservation–Path Factorial Main Effects and Interaction

Exact values behind Figure S4, recomputed from the exploratory seven-series factorial (`factorial_interactions.json`). The family was fixed for the v3 diagnostic run but not before source access, so every entry stays exploratory. Intervals are 95% series-cluster bootstrap intervals; the exact column is the series sign-flip value and the final column its Holm adjustment inside the six-contrast family. Mean Δ is in ROUGE-L F1.

Budget	Factor	Mean Δ	95% CI low	95% CI high	Paired SMD	Exact p	Holm p
110	path channel (paths off vs on)	-0.0025	-0.0179	+0.0133	-0.109	0.7969	1.0000
110	role-group reservation (on vs off)	-0.0097	-0.0175	-0.0024	-0.882	0.0469	0.2344
110	reservation x path interaction	-0.0110	-0.0317	+0.0026	-0.413	0.3438	1.0000
260	path channel (paths off vs on)	-0.0103	-0.0187	-0.0037	-0.930	0.0312	0.1875
260	role-group reservation (on vs off)	+0.0005	-0.0039	+0.0053	+0.068	0.8750	1.0000
260	reservation x path interaction	+0.0063	-0.0022	+0.0160	+0.478	0.2500	1.0000

Per-series deltas for the same six contrasts are supplied as `q5_reservation_path_series_deltas.csv`. Neither main effect survives family-wise correction at either budget.

Table S15. External Prospective Corpus Stratified by Publisher Family

Paired ROUGE-L F1 differences inside each publisher family of the frozen 24-report external corpus. Families are reported separately and are never pooled into a single accuracy; family sizes are 2–9, so every entry is descriptive and no test is applied. Candidate units and reference words are given as the observed range inside the family.

Family	Budget	Contrast	Reports	Candidates	Reference words	Mean Δ	neg/pos/tie
ENTSO-E frequency-deviation reports	word:110	AB2 - AB0	2	271–336	401–800	-0.0834	2/0/0
ENTSO-E frequency-deviation reports	word:110	Full - no_path	2	271–336	401–800	+0.0085	0/2/0
ENTSO-E frequency-deviation reports	word:110	TextRank - no_path	2	271–336	401–800	-0.0105	1/1/0
ENTSO-E frequency-deviation reports	word:260	AB2 - AB0	2	271–336	401–800	-0.0235	1/1/0
ENTSO-E frequency-deviation reports	word:260	Full - no_path	2	271–336	401–800	-0.0799	2/0/0
ENTSO-E frequency-deviation reports	word:260	TextRank - no_path	2	271–336	401–800	-0.0893	1/1/0
ENTSO-E grid incidents	word:110	AB2 - AB0	7	589–2000	1190–10935	-0.0026	5/2/0
ENTSO-E grid incidents	word:110	Full - no_path	7	589–2000	1190–10935	-0.0010	2/3/2
ENTSO-E grid incidents	word:110	TextRank - no_path	7	589–2000	1190–10935	+0.0205	0/7/0
ENTSO-E grid incidents	word:260	AB2 - AB0	7	589–2000	1190–10935	+0.0198	2/5/0
ENTSO-E grid incidents	word:260	Full - no_path	7	589–2000	1190–10935	-0.0028	5/0/2
ENTSO-E grid incidents	word:260	TextRank - no_path	7	589–2000	1190–10935	+0.0175	1/6/0
ENTSO-E market-coupling incidents	word:110	AB2 - AB0	9	21–84	108–779	-0.0652	7/1/1
ENTSO-E market-coupling incidents	word:110	Full - no_path	9	21–84	108–779	-0.0012	2/0/7
ENTSO-E market-coupling incidents	word:110	TextRank - no_path	9	21–84	108–779	+0.0523	3/6/0
ENTSO-E market-coupling incidents	word:260	AB2 - AB0	9	21–84	108–779	-0.0614	5/3/1
ENTSO-E market-coupling incidents	word:260	Full - no_path	9	21–84	108–779	+0.0036	1/1/7
ENTSO-E market-coupling incidents	word:260	TextRank - no_path	9	21–84	108–779	+0.0604	3/6/0
NERC event reports	word:110	AB2 - AB0	6	37–334	102–1431	-0.0118	4/1/1
NERC event reports	word:110	Full - no_path	6	37–334	102–1431	-0.0060	2/0/4
NERC event reports	word:110	TextRank - no_path	6	37–334	102–1431	-0.0082	3/3/0
NERC event reports	word:260	AB2 - AB0	6	37–334	102–1431	+0.0058	3/3/0
NERC event reports	word:260	Full - no_path	6	37–334	102–1431	-0.0031	2/0/4
NERC event reports	word:260	TextRank - no_path	6	37–334	102–1431	+0.0073	2/4/0

The 260-word path-layer difference observed on the pooled corpus is carried by the two frequency-deviation reports; the grid-incident, market-coupling and NERC families sit close to zero. Per-condition family means are supplied as `q9_external_family_condition_means.csv`.

Table S16. Per-Report Cost Against Candidate-Set Size, and Role/Edge Coverage

Measured cost and coverage for the seven post-access external series. Candidate counts come from the private derived corpus and are reported as counts only; times are selection and scoring seconds averaged over the thirteen conditions at that budget, excluding the separately loaded embedding model. Role and typed-edge coverage are formal channel coverage, not accuracy against human judgement.

Series	Candidates	Budget	Selection (s)	Peak MB	Role coverage	Typed-edge coverage
series_entsoe_czech_2025	695	110	0.082	1.24	0.846	0.923
series_entsoe_czech_2025	695	260	0.170	1.43	0.872	0.978
series_entsoe_mepso_2025	734	110	0.101	1.29	0.821	1.000
series_entsoe_mepso_2025	734	260	0.203	1.67	0.872	0.985
series_entsoe_see_2024	1408	110	0.120	2.47	0.872	1.000
series_entsoe_see_2024	1408	260	0.263	2.94	0.872	1.000
series_nerc_gmd_2024	200	110	0.054	0.40	0.897	0.936
series_nerc_gmd_2024	200	260	0.087	0.46	0.949	0.947
series_nerc_large_load_loss	58	110	0.042	0.23	0.846	0.931
series_nerc_large_load_loss	58	260	0.064	0.49	0.872	0.949
series_nerc_low_wind	153	110	0.039	0.25	0.923	0.866
series_nerc_low_wind	153	260	0.090	0.44	0.923	0.922
series_transpower_tower_130	830	110	0.108	1.60	0.821	0.729
series_transpower_tower_130	830	260	0.237	1.86	0.897	0.814

Per-condition values, including redundancy and budget utilization, are supplied as `q4_role_edge_coverage_per_condition.csv`; the four-condition cost detail is in `q3_report_cost_vs_candidates.csv`.

Table S17. One-Sided Bounds on the Frozen External Layer Contrasts

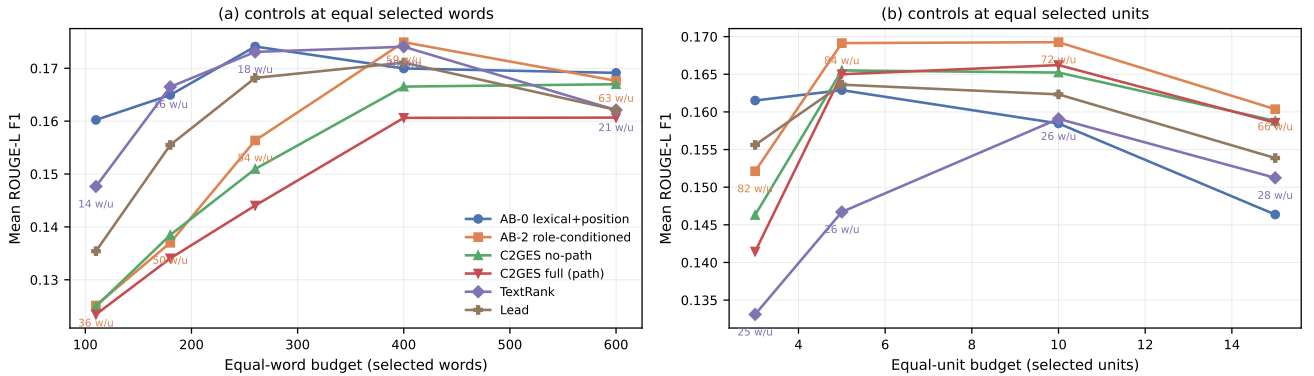
Post hoc bounds computed on the same frozen 24-report record as Table S10 and the arm-level experiment. The upper limit is the one-sided 95% bootstrap percentile of the paired mean difference (200 000 resamples, seed 20260926), i.e. the largest gain still compatible with the data. The leave-one-out columns report the span of the mean and the worst-case upper limit when any single report is removed. Bounds are post hoc and are not pre-registered confirmatory tests.

Contrast	Budget	Reports	Mean Δ	neg/pos/tie	Upper limit (95%)	LOO mean range	LOO worst limit
path layer (Full – no-path)	unit:10	24	+0.0010	8/6/10	+0.0038	[+0.0001, +0.0016]	+0.0043
path layer (Full – no-path)	unit:5	24	-0.0005	5/4/15	+0.0047	[-0.0026, +0.0012]	+0.0059
path layer (Full – no-path)	word:110	24	-0.0015	6/5/13	+0.0006	[-0.0020, -0.0004]	+0.0012
path layer (Full – no-path)	word:260	24	-0.0069	10/1/13	+0.0021	[-0.0094, -0.0006]	+0.0038
role layer (AB-2 – AB-0)	unit:10	24	+0.0108	9/15/0	+0.0228	[+0.0074, +0.0146]	+0.0255
role layer (AB-2 – AB-0)	unit:5	24	+0.0063	10/14/0	+0.0180	[+0.0036, +0.0126]	+0.0209
role layer (AB-2 – AB-0)	word:110	24	-0.0351	18/4/2	-0.0166	[-0.0385, -0.0287]	-0.0125
role layer (AB-2 – AB-0)	word:260	24	-0.0178	11/12/1	+0.0033	[-0.0207, -0.0074]	+0.0077
TextRank baseline	word:110	24	+0.0227	7/17/0	+0.0442	[+0.0140, +0.0280]	+0.0492
TextRank baseline	word:260	24	+0.0221	7/17/0	+0.0440	[+0.0154, +0.0316]	+0.0503
Lead baseline	word:110	24	+0.0104	9/15/0	+0.0276	[+0.0038, +0.0133]	+0.0306
Lead baseline	word:260	24	+0.0172	10/14/0	+0.0372	[+0.0092, +0.0209]	+0.0407

The path-layer upper limit stays at or below +0.0047 in every budget and at or below +0.0059 under leave-one-out, so a path-channel gain larger than +0.005 ROUGE-L is excluded on this corpus even though the mean-based test is not significant. The role-layer upper limit is positive under equal-unit budgets and negative at 110 words, which is the budget dependence reported in the main text.

Figure S5. Budget-Efficiency Curves on the Frozen External Corpus

Arm ranking on the frozen 24-report external corpus depends on the budget type (labels: selected words per unit)



Panel (a) holds the selected-word budget fixed (110–600 words) and panel (b) the selected-unit budget (3–15 units); labels give the realised selected words per unit. The ordering of the arms is a property of the budget type: TextRank and Lead lead at mid-range word budgets, whereas the role-conditioned arms lead under equal-unit budgets while packing roughly 1.5 times as many words per selected unit as the lexical-and-position baseline and about three times as many as TextRank. Source values are supplied as `figS5_source.csv` and the full arm record as `external_arms_curve_v4.json`.

Table S18. Out-of-Domain Transfer Check on GovReport

The frozen arm set applied without tuning to 100 length-stratified documents from the GovReport test split (CRS and GAO reports with human-written summaries, CC BY 4.0; drawn with seed 20260926 from 973 test rows). This is a robustness transfer layer, not domain evidence: the role cue lexicon is power-domain specific (mean formal role coverage 0.246, minimum 0.016) and the references are abstractive. Intervals and limits are computed as in Table S17 (200 000 bootstrap samples).

Contrast	Budget	Reports	Mean Δ	neg/pos/tie	Upper limit (95%)	LOO mean range	LOO worst limit
path layer (Full – no-path)	word:110	100	-0.0031	24/15/61	-0.0011	[-0.0033, -0.0019]	-0.0006
path layer (Full – no-path)	word:260	100	-0.0026	37/25/38	-0.0007	[-0.0031, -0.0022]	-0.0004
path layer (Full – no-path)	unit:5	100	-0.0052	39/13/48	-0.0030	[-0.0055, -0.0044]	-0.0025
path layer (Full – no-path)	unit:10	100	-0.0028	30/21/49	-0.0010	[-0.0030, -0.0024]	-0.0007
role layer (AB-2 – AB-0)	word:110	100	-0.0212	81/17/2	-0.0161	[-0.0220, -0.0200]	-0.0149
role layer (AB-2 – AB-0)	word:260	100	-0.0204	77/22/1	-0.0143	[-0.0218, -0.0195]	-0.0135
role layer (AB-2 – AB-0)	unit:5	100	-0.0007	52/48/0	+0.0062	[-0.0028, +0.0006]	+0.0072
role layer (AB-2 – AB-0)	unit:10	100	-0.0098	71/29/0	-0.0038	[-0.0111, -0.0087]	-0.0030
TextRank – no-path	word:110	100	+0.0215	15/84/1	+0.0264	[+0.0209, +0.0230]	+0.0274
TextRank – no-path	word:260	100	+0.0273	14/86/0	+0.0332	[+0.0264, +0.0287]	+0.0342
TextRank – no-path	unit:5	100	+0.0005	43/57/0	+0.0072	[-0.0003, +0.0029]	+0.0085
TextRank – no-path	unit:10	100	+0.0119	33/67/0	+0.0182	[+0.0109, +0.0138]	+0.0194
Lead – no-path	word:110	100	+0.0034	57/42/1	+0.0123	[+0.0007, +0.0043]	+0.0132
Lead – no-path	word:260	100	+0.0044	62/38/0	+0.0170	[-0.0005, +0.0056]	+0.0182
Lead – no-path	unit:5	100	-0.0250	81/19/0	-0.0113	[-0.0308, -0.0238]	-0.0101
Lead – no-path	unit:10	100	-0.0166	74/26/0	-0.0038	[-0.0203, -0.0153]	-0.0024

Every path-layer upper limit is below zero, so a positive path-channel effect is excluded on this corpus at all four budgets, and the role-conditioned arms are significantly worse than their lexical baseline at both word budgets. Per-document records are supplied as `govreport_transfer_v1.json`.

Sensitivity: the frozen strata fully sampled (970 of 973 test rows). The same frozen protocol run over every document of the five strata, as a sensitivity rather than a new pre-registered test. Holm values are those of the full arm-by-budget family (twelve contrasts).

Contrast	Budget	Reports	Mean Δ	neg/pos/tie	Upper limit (95%)
path layer (Full – no-path)	word:110	970	-0.0021	233/156/581	-0.0013
path layer (Full – no-path)	word:260	970	-0.0017	330/224/416	-0.0008
path layer (Full – no-path)	unit:5	970	-0.0032	316/166/488	-0.0024
path layer (Full – no-path)	unit:10	970	-0.0020	306/197/467	-0.0013
role layer (AB-2 – AB-0)	word:110	970	-0.0193	722/233/15	-0.0175
role layer (AB-2 – AB-0)	word:260	970	-0.0220	725/241/4	-0.0199
role layer (AB-2 – AB-0)	unit:5	970	-0.0030	508/460/2	-0.0009
role layer (AB-2 – AB-0)	unit:10	970	-0.0134	632/337/1	-0.0112

Every path-layer upper limit remains below zero at $n = 970$ (worst case -0.0008), and the path contrasts are family-corrected significant at all four budgets (Holm 0.0002–0.0040); the role layer is negative at every budget as well. Pairwise records are supplied as `govreport_transfer_full970_pairs.csv`.

Energy-topic subset of the same split (160 documents). Selecting reports whose body matches at least three entries of a frozen keyword list (energy, grid, transmission, power plant, electricity, renewable, nuclear, and so on; the rule was fixed before the subset was formed) gives a topic stratum that is closer to the paper’s domain than the full split. The path-layer differences at the four budgets are $-0.0009 / -0.0016 / -0.0018 / -0.0025$, every positive share of discordant documents is at or below 41%, and the worst one-sided 95% upper limit is $+0.0008$, so the $+0.005$ margin still holds; the ten-unit contrast is family-corrected significant (Holm 0.0174) while the word-budget contrasts are not. Records: `govreport_energy160_pairs.csv`.

Table S19. Reference-Type Sensitivity with Human-Written Highlights

The frozen arm set applied to 400 length-stratified CNN/DailyMail test articles whose reference is the human-written highlight bullet points rather than an abstractive summary. This is a reference-type sensitivity, not domain evidence: the corpus is news, the articles are short (median 23 candidate units), and the power-domain cue lexicon barely applies (mean formal role coverage 0.116). The point of the layer is that the path channel leaves the selection untouched in almost every document, so a null cannot be an artefact of comparing extracts with rewritten text.

Contrast	Budget	Articles	Mean Δ	neg/pos/tie	Upper limit (95%)
path layer (Full – no-path)	word:110	400	+0.0005	14/10/376	+0.0016
path layer (Full – no-path)	word:260	400	-0.0003	18/14/368	+0.0002
path layer (Full – no-path)	unit:5	400	-0.0000	4/4/392	+0.0002
path layer (Full – no-path)	unit:10	400	+0.0000	1/4/395	+0.0000
role layer (AB-2 – AB-0)	word:110	400	-0.0092	187/135/78	-0.0055
role layer (AB-2 – AB-0)	word:260	400	-0.0018	134/119/147	-0.0005
role layer (AB-2 – AB-0)	unit:5	400	-0.0065	180/131/89	-0.0034
role layer (AB-2 – AB-0)	unit:10	400	-0.0013	125/130/145	+0.0003

The path-layer difference is zero to the fourth decimal at three of the four budgets, with 92–99% of articles exactly tied, and every one-sided upper limit is at most +0.0016 — far inside the +0.005 margin. Under this reference the term is inert rather than harmful, and the position baseline dominates (Lead’s advantage is family-corrected significant at all four budgets). Per-document pairs are supplied as `reference_type_400_pairs.csv`.

Table S20. Frozen Parent–Held-Out Synthetic Stress Fixtures

Sixteen fictional reports in eight series generated under the frozen protocol `PROTOCOL_synthetic_c2ges_gendata_v1.md` (sha256 prefix `daf7c969`, thresholds unchanged after freezing) and shipped with the run manifests, gate ledgers, dataset card and the component-factorial records. The parent and held-out sets use disjoint themes, and the two generator families are swapped between them. All nineteen deterministic gates pass in both sets; the fixtures are software and endpoint stress material and never enter the E1/E2/E3 evidence chain.

Set	Method	110 words	260 words	Set	Method	110 words	260 words
Parent	Full C ² GES	0.1302	0.1184	Held-out	Full C ² GES	0.1168	0.1082
Parent	No-path C ² GES	0.1199	0.1086	Held-out	No-path C ² GES	0.1171	0.1039
Parent	G-T (typed, no path)	0.1199	0.1086	Held-out	G-T (typed, no path)	0.1171	0.1039
Parent	G-U (untyped, no path)	0.0887	0.0981	Held-out	G-U (untyped, no path)	0.0826	0.0910

Full–no-path macro-mean differences (110/260 words) match the table: parent +0.0103 / +0.0098; held-out -0.0003 / +0.0043. Series-equal Full–no-path (110/260 words): parent Full–no-path +0.0147 / +0.0178 (Holm 0.75 / 0.75); held-out Full–no-path -0.0007 / +0.0032 (Holm 1.00 / 1.00). Typed-minus-untyped graph: parent G-T–G-U +0.0312 / +0.0104 (Holm 0.25 / 0.25); held-out G-T–G-U +0.0344 / +0.0129 (Holm 0.25 / 0.25). Reservation main effect: parent reservation +0.0089 / -0.0098 (Holm 1.00 / 1.00); held-out reservation +0.0040 / +0.0111 (Holm 1.00 / 1.00). No corrected path contrast is nonzero in either set.

Gate	Frozen rule (abridged)	Parent measured	Held-out measured	Pass
S1	4 series x 2 reports each	8/8	8/8	19/19
S2	reference equals the concatenation of reference_unit_ids	8/8	8/8	19/19
S3	causal DAG acyclic	8/8	8/8	19/19
S4	semantic core 18–26 sentences; 10–28 words	8/8	8/8	19/19
S5	five roles x 2–6 each, non-uniform; one priority=2 per role	8/8	8/8	19/19
S6	every report covers all five roles	8/8	8/8	19/19
S7	Synthetic facility prefix; forbidden tokens zero	8/8	8/8	19/19
S8	24–36 distinct distractors; 10–24 words	8/8	8/8	19/19
D1	candidate-count W1 $j=0.75$	0.208	0.208	19/19
D2	page-count W1 $j=0.75$	0.131	0.131	19/19
D3	layout-share max abs(delta) $j=0.03$	0.000	0.000	19/19
D4	exact duplicate rate $j=0.20$	0.000	0.000	19/19
D5	unit-length W1 $j=0.25$	0.189	0.174	19/19
D7	core-sentence first-quartile share $j=0.25$	0.000–0.250	0.000–0.238	19/19
D8	max Jaccard vs real corpus $j=0.15$; hit share $j=2\%$	0.120 / 0.000	0.062 / 0.000	19/19
D9	distractor families $i=6$; single family $j=40\%$	150–675 / 0.011	150–675 / 0.028	19/19
E1b	within-set per-report AUROC $i=0.80$	0.894–0.970	0.875–0.945	19/19
E2	distinct-2 $i=0.526$	0.452–0.656	0.535–0.656	19/19
E3	vocabulary overlap $i=0.55$	0.675	0.627	19/19

The leakage gate D8 keeps every unit’s maximum Jaccard overlap with the real corpus at or below 0.12 (parent) / 0.06 (held-out) against a frozen 0.15 bound; E1b is the within-set per-report document-identity AUROC (frozen floor 0.80), E2 is distinct-2 and E3 is vocabulary overlap with the development lexicon. AUROC(synthetic-vs-real) is reported but descriptive only: it measures document identity, not realism. The fixture is a synthetic assembly with visible seams at distractor joints, an easily separable document identity, and only eight reports per set, so it supports software and mechanism diagnosis but not statistical inference; the parent line saw generator-side tuning on intermediate gate results (thresholds unchanged) and the held-out line is the independent check (README and dataset card shipped beside the records). Records: `synthetic_stress_v1/run_20260927_gendata_parent_v3r1/` and `synthetic_stress_v1/run_20260927_gendata_heldout_v2r3/`.

Exploratory Development-Only Calibration

The complete chronology, configuration space, leave-one-report-out results, and machine-readable records are described in `03_Reproducibility/Data/POST_UNBLINDING_DEV_CALIBRATION_SUPPLEMENT.md`. A zero path-deletion weight was selected in all 12 development folds; the analysis was not applied to the retained 15-report test set.

A separate equal-budget exercise is documented in `03_Reproducibility/Data/dev_balanced_tuning/`. Semantic-MMR, TextRank, and normalized-path C²GES each received nine configurations and the same ordered development objective over both extraction budgets. The selected values were 0.9, 0.65, and 0.0, respectively. The decision record states `test_input_accessed=false`; these configurations are authorized only for a future preregistered external-series comparison and do not alter the retained-test results.